

Splat: Developing a "Strange" Shader

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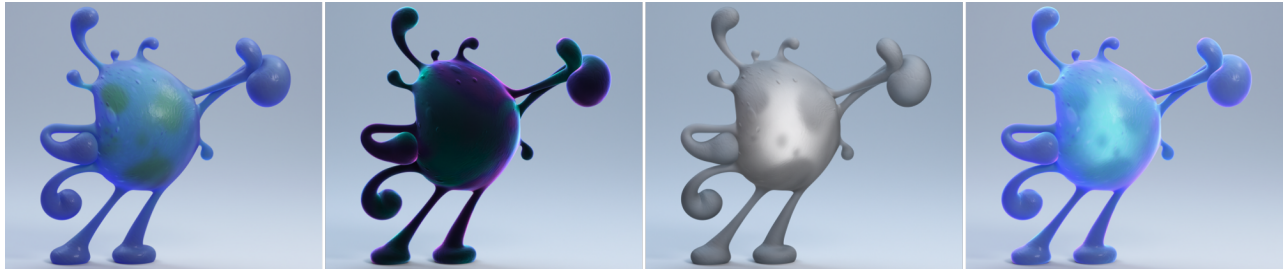


Figure 1: Shader components, left to right: (a) basic surface shader, (b) chromatic edge, (c) faux volume emission, (d) final shader. ©2023 Disney

ABSTRACT

In Walt Disney Animation Studios’ “Strange World”, the character Splat presented unique artistic requirements that were computationally prohibitive and unwieldy to achieve using previous techniques at feature film scale. In this paper, we present our solution to two of these challenges: the chromatic edge along the outer rim of the character, and an efficient, deforming heterogeneous volume for the inner glow within Splat. Importantly, we were able to achieve these effects in a single render pass, greatly simplifying our workflow while achieving the desired artistic direction.

ACM Reference Format:

Kendall Litaker, Dan Lipson, Mason Khoo, and Brent Burley. 2023. Splat: Developing a “Strange” Shader. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Talks (SIGGRAPH ’23 Talks)*, August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3587421.3595424>

1 INTRODUCTION

Our artists faced the challenging task of creating Splat, a blobby creature with the amorphous shape of an ameoba, and with a gummy, translucent surface surrounding floating organelles like a jellyfish. In addition to having a bioluminescent inner glow, Splat’s design incorporated a chromatic edge effect inspired by the chromatic aberrations caused when light refracts through tiny creatures

under a microscope. Because Splat communicates largely with pantomime, stretching and squashing into any number of shapes, our look would need to hold up under extreme deformation.

2 EARLY EXPLORATIONS

Our first attempt used simple translucency controlled with facing ratio to create an inner glow with a defined edge (Figure 2, top). This method produced excellent render speed and noise levels, but required constant backlighting in order to maintain saturation levels. Using external lights as the source of the internal glow also tended to shift Splat’s perceived color towards the lights’ colors and away from the desired teal color. Adding character-specific lights in addition to normal scene lighting required impractical amounts of micromanagement. For the chromatic edge, using camera-based facing ratio also produced a problematically uniform border around all of Splat regardless of scene lighting (Figure 2, bottom left), making it difficult to integrate Splat with the surrounding environment.

To introduce more art directability and natural lighting to the chromatic edge, we tried combining the original facing ratio strategy with a camera projection of a separately rendered pre-pass of Splat lit with a gray diffuse shader. This approach altered the brightness of the emission on the edge based on the lighting in the scene, creating natural shadows to help situate the character into the space (Figure 2, bottom right); however, using a separately rendered pre-pass is inefficient and complicated our workflows. The effect would not appear in renders prior to lighting, which would make judging the character’s look difficult in upstream departments.

For our next attempt for the inner glow, we used heterogeneous volumes. They provided the most artistic control, but had a number of drawbacks; they were inefficient and noisy to render, they required the FX department to regenerate a VDB file per frame any time Splat’s animation was updated, and they utilized a separate

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SIGGRAPH ’23 Talks, August 06–10, 2023, Los Angeles, CA, USA

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ACM ISBN 979-8-4007-0143-6/23/08.

<https://doi.org/10.1145/3587421.3595424>

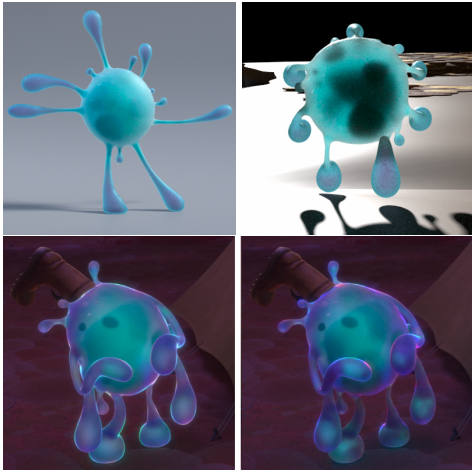


Figure 2: Early tests using backlighting (top), and using facing ratio, before and after diffuse camera projection (bottom) ©2023 Disney

render pass which is difficult to manage. Additionally, some poses exhibited harsh discontinuities where the limbs joined to the body (Figure 3, top). We tried treating the space between the limb and the body as part of the volume, which helped resolve the artifact, but it also altered the overall translucent look, and having to manage this selectively did not scale well from a workflow perspective.

To address these concerns, a small team of technical directors, software engineers, lighters, and look artists investigated more lightweight and efficient solutions. Our goal was to create an all-in-one shader that would achieve the inner glow and chromatic edge in an art-directable, efficient, single pass approach.

3 CHROMATIC EDGE

For the chromatic edge, we adopted a new multiple-scattering sheen approach [Zeltner et al. 2022] that simulates a thin layer of particles/fibers on a surface, producing a more naturalistic result in a variety of shot lighting scenarios, and eliminating the separate render pass of our previous approach. We added further variation to the chromatic edge by introducing a slowly animating procedural noise into the rainbow colors. In order to exaggerate the effect, we scaled the chromatic edge to amplify the incident light (Figure 1b) beyond what would normally be reflected. We added a control to break out the sheen response into a separate render output, allowing lighters to make further adjustments during compositing.

4 HETEROGENEOUS VOLUME

To avoid the overhead of per-frame VDB volumes inside of Splat, we developed a novel “faux volume emission” technique (Figure 1c). As in the full volume method, we accumulated volumetric emission along each ray traced within Splat. We shaped the volumetric emission density through expressions using region grads [Burley et al. 2018] specified in Splat’s rest pose. In order for the volume to track with character animation, we transformed the surface points and ray directions entering Splat into reference space using PRef vertex attributes before evaluating the density expression. For efficiency, we sampled the density at a single randomly chosen point along each ray. Even with this new shader, the limb discontinuity issue

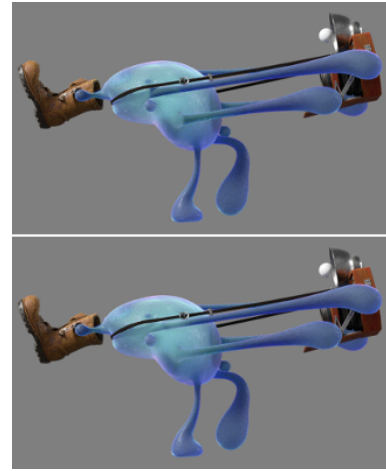


Figure 3: Limb discontinuity before and after fix. ©2023 Disney

was still present. The arms were thin and thus accumulated little volumetric emission and appeared fairly dark. Where the arm was sharply folded, the transition from the dark arm to the much lighter body appeared very abrupt. Though physically correct, this visual discontinuity was artistically undesirable. To address this, we added a control to allow some emission to bleed through from the body to the arm proportional to how close the arm is to the body (Figure 3, bottom).

5 RESULTS

We encountered some issues during initial production testing of our new shader. The faux volume calculation depends on a ray entering and exiting an object to accumulate light; any intersection from another object into Splat would cause the emission to diminish in brightness in an undesirable way. Lighters addressed this in problematic shots by manually tagging the intersecting objects to be ignored when calculating the faux volume emission. The chromatic edge also needed to be adjusted in some shots in order to create better contact between Splat’s “feet” and the ground; lighters used region grads to give the chromatic edge a soft falloff near areas of contact.

The faux volume shader successfully eliminated the need for the proof-of-concept shader’s heaviest component, the per-frame VDB volume, and resulted in images with lower noise levels and render times. Additional work from the FX department was no longer necessary, which streamlined our workflow. This approach also ensured that work-in-progress renders from upstream departments accurately represented Splat’s final look. Once in lighting, only limited adjustments were needed to finalize Splat’s look to get creative approval.

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